

**BEFORE YOU READ — ANSWER FROM MEMORY**

1. Which event in infancy permanently changes oral colonisation, and why?
2. Put the stages of dental plaque formation in order (five steps).
3. What one feature separates the pharyngeal membrane of diphtheria from oral thrush?

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

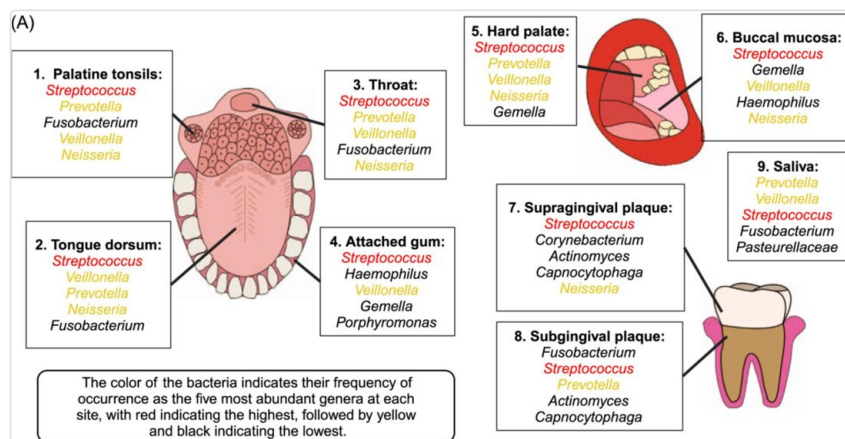
1 The mouth as an ecosystem

- Connects the **external environment** with the respiratory and GI tracts; **second-largest microbiota** in the body
- **>700 bacterial species** plus fungi, protozoa, archaea, viruses; organisms **co-aggregate and compete** rather than live alone
- Most are **anaerobic or facultatively anaerobic**; temperature, O₂, pH, saliva flow and diet vary daily → many microhabitats within centimetres

Niche (lecture figure)	Most abundant	Next	Lowest of the top five
1 Palatine tonsils	<i>Streptococcus</i>	<i>Prevotella</i> , <i>Veillonella</i> , <i>Neisseria</i>	<i>Fusobacterium</i>
2 Tongue dorsum	<i>Streptococcus</i>	<i>Veillonella</i> , <i>Prevotella</i> , <i>Neisseria</i>	<i>Fusobacterium</i>
3 Throat	<i>Streptococcus</i>	<i>Prevotella</i> , <i>Veillonella</i> , <i>Neisseria</i>	<i>Fusobacterium</i>
4 Attached gum	<i>Streptococcus</i>	<i>Veillonella</i>	<i>Haemophilus</i> , <i>Gemella</i> , <i>Porphyromonas</i>
5 Hard palate	<i>Streptococcus</i>	<i>Prevotella</i> , <i>Veillonella</i> , <i>Neisseria</i>	<i>Gemella</i>
6 Buccal mucosa	<i>Streptococcus</i>	<i>Veillonella</i> , <i>Neisseria</i>	<i>Gemella</i> , <i>Haemophilus</i>
7 Supragingival plaque	<i>Streptococcus</i>	<i>Neisseria</i>	<i>Corynebacterium</i> , <i>Actinomyces</i> , <i>Capnocytophaga</i>
8 Subgingival plaque	<i>Streptococcus</i>	<i>Prevotella</i>	<i>Fusobacterium</i> , <i>Actinomyces</i> , <i>Capnocytophaga</i>
9 Saliva	<i>Streptococcus</i>	<i>Prevotella</i> , <i>Veillonella</i>	<i>Fusobacterium</i> , <i>Pasteurellaceae</i>

Colour code in the source figure: red = most frequent, yellow = next, black = lowest of the five most abundant genera per site.

***Streptococcus* is red at every site.**



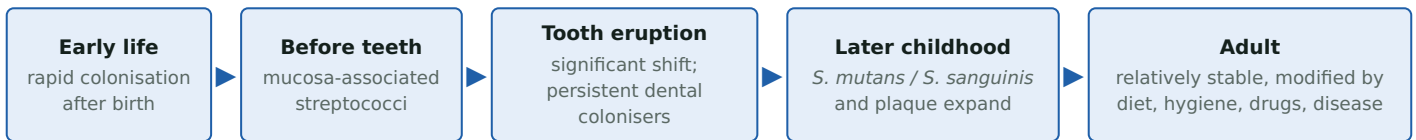
Nine oral microenvironments and their five most abundant genera (lecture figure).

2 Who lives there

Group	Key points
Resident	Present in all humans; mostly permanent; commensal or mutualistic ; give colonisation resistance and stability; can turn opportunistic
Transient	From food, water, hands, contacts; removed by resident competitors, immune defences and saliva flow ; may infect if barriers are compromised

Core vs variable	Core = taxa shared by healthy people; variable = differs between people and over time; both shaped by age, diet, habits, medication
Gram-positive	Viridans streptococci (<i>S. mutans</i> , <i>mitis</i> , <i>oralis</i> , <i>salivarius</i> , <i>sanguinis</i>); <i>Actinomyces</i> ; <i>Peptostreptococcus</i>
Gram-negative / anaerobes	<i>Veillonella</i> , <i>Prevotella</i> , <i>Porphyromonas</i> , <i>Fusobacterium</i> , <i>Capnocytophaga</i> , <i>Neisseria</i>
Others	<i>Candida</i> may be carried without disease; viruses may persist or reactivate in oral / tonsillar tissue. Presence ≠ disease

3 Establishment across life — tooth eruption is the key transition

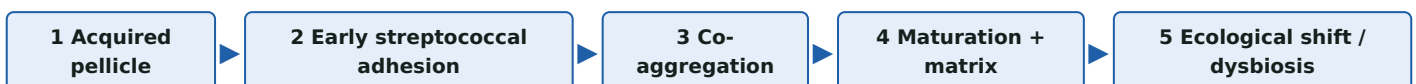


Teeth create a permanent, non-shedding surface where plaque can accumulate.

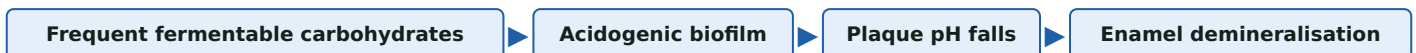
4 What shapes the community

Factor	Effect
Saliva — physical	Continuous flow dilutes and removes microbes and distributes them between surfaces
Saliva — antimicrobial	Secretory IgA, lysozyme, peroxidase system, antimicrobial peptides
Nutrients	Saliva is nutrient-poor, but food debris, epithelial material and gingival fluid make rich local sites → growth on teeth, tongue, gingival margins
pH	Acidification after carbohydrate intake
Oxygen	Highest on exposed surfaces; lowest in gingival crevices
Diet	Frequency and fermentable carbohydrates matter
Hygiene / host immunity	Mechanical disruption of plaque / mucosal and systemic defences
Drugs / age & habits	Antibiotics, steroids disturb balance / tooth status, smoking, dentures, xerostomia

5 Plaque biofilm and caries



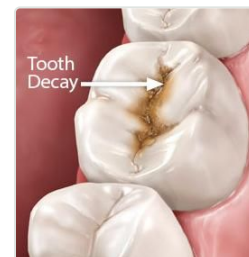
Biofilm communities sit in an extracellular matrix: cooperative, persistent, less susceptible to host defences.



- ***S. mutans* group** = major caries-associated streptococci and important plaque producers; ***Lactobacillus*** adds acid production and progression in an established acidic environment
- **Caries and periodontal disease are biofilm-associated ecological diseases — not single-organism infections**



Dental plaque on teeth.



Tooth decay.

6 Beyond the teeth

Concept	Point
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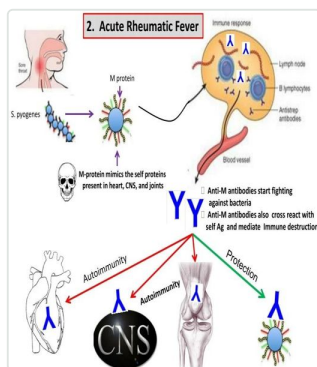
Local extension	Periodontal and dental infections may spread to jaw, sinus, facial spaces
Bloodstream access	Mucosal trauma or dental procedures → transient bacteraemia
Endocarditis link	Viridans streptococci = classic oral organisms in subacute endocarditis
Immunocompromised host	Oral <i>Candida</i> and invasive fungal lesions become more important
Oropharyngeal gateway	Tonsils and pharyngeal mucosa: bacterial and viral infection and transmission

7 Pharyngitis — think syndrome first

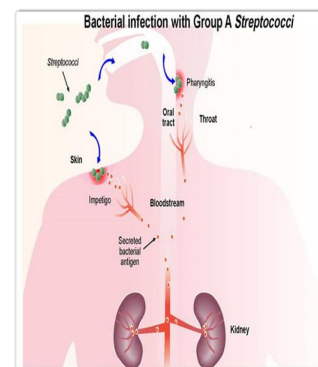
Bacterial	Viral	Fungal / other
<i>Streptococcus pyogenes</i> (GAS); <i>S. dysgalactiae</i> ; <i>Fusobacterium necrophorum</i> ; <i>Corynebacterium diphtheriae</i> ; <i>Neisseria gonorrhoeae</i>	EBV, adenovirus, influenza / parainfluenza, rhinovirus / coronavirus, Coxsackie A, HSV	<i>Candida</i> (immune suppression, local risk factors); ulcerative fungal disease in selected immunocompromised patients

GAS is the key bacterial diagnosis; aim to spot the few diagnoses that change management.

Complication of GAS	Key points
Suppurative	Peritonsillar infection / abscess, cervical lymphadenitis, otitis / sinus extension, deep neck infection
Acute rheumatic fever	Immune-mediated ; heart, joints, skin, CNS; M-protein mimics self proteins (figure); timely treatment of GAS pharyngitis reduces risk
Acute glomerulonephritis	Immune-mediated ; may follow pharyngeal or skin infection; oedema, hypertension, haematuria, proteinuria



Acute rheumatic fever — M-protein mimicry.



Group A streptococcal infection → glomerulonephritis.

8 Discriminator grid — sore throat and mouth look-alikes

Condition	Organism	ONE finding that separates it
GAS pharyngitis	<i>S. pyogenes</i>	The key bacterial cause; risk of rheumatic fever / glomerulonephritis
Diphtheria	Toxin-producing <i>C. diphtheriae</i>	Adherent gray pseudomembrane , "bull neck"; isolate, notify, urgent management
Gonococcal pharyngitis	<i>N. gonorrhoeae</i>	Relevant sexual exposure ; may be mild or silent; site-appropriate testing
Oral thrush	<i>Candida</i>	White plaques that can often be removed , leaving an erythematous surface; risk: immunosuppression, antibiotics, steroids, diabetes, dentures, xerostomia
Herpes (HSV)	HSV	Painful vesicles / ulcers ; recurs after latency
Measles	Measles virus	Koplik spots on buccal mucosa



Diphtheria membrane.



Gonococcal pharyngitis.



Oral thrush.



Koplik spots.

P Must-remember pearls

- Oral microbiota are **place-specific**; ***Streptococcus*** is the top genus at every site in the figure.
- **Tooth eruption** = permanent, non-shedding surface → plaque biofilm; plaque steps: pellicle → streptococcal adhesion → co-aggregation → matrix → dysbiosis.
- Caries = **fermentable carbohydrates** → **acidogenic biofilm** → **pH falls** → **demineralisation** (*S. mutans*, *Lactobacillus*); an ecological disease.
- Viridans streptococci + dental procedures/trauma → **subacute endocarditis**.
- **GAS** is the key bacterial pharyngitis → rheumatic fever / glomerulonephritis; **diphtheria** = **adherent gray membrane**, **thrush** = **removable white plaques**.

Q Test yourself

1. A toddler's first teeth erupt. Which change best explains the shift in oral microbiota?

- | | |
|------------------------------|---|
| A. Saliva becomes sterile | B. A permanent, non-shedding surface allows persistent plaque |
| C. The mouth becomes aerobic | D. Lactobacilli disappear |

2. A teenager sips sugary drinks all day and develops caries. Which sequence is correct?

- | | |
|---|---------------------------------------|
| A. Acidogenic biofilm → plaque pH falls → enamel demineralisation | B. Plaque pH rises → remineralisation |
| C. Salivary IgA falls → caries | D. Diphtheria toxin → enamel loss |

3. A child has a sore throat, an adherent gray pharyngeal membrane and a swollen neck. Which organism?

- | | |
|---------------------|---------------------------------------|
| A. <i>Candida</i> | B. <i>Corynebacterium diphtheriae</i> |
| C. <i>S. mutans</i> | D. <i>Neisseria gonorrhoeae</i> |

4. A denture wearer on antibiotics has white oral plaques that scrape off leaving red mucosa. Diagnosis?

- | | |
|----------------|--------------------|
| A. Oral thrush | B. Koplik spots |
| C. Diphtheria | D. Herpetic ulcers |

5. Untreated GAS pharyngitis is followed by joint pain and carditis. Which complication?

- | | |
|-----------------------------|--------------------------|
| A. Acute glomerulonephritis | B. Acute rheumatic fever |
| C. Peritonsillar abscess | D. Subacute endocarditis |

ANSWERS — COVER UNTIL YOU'VE COMMITTED

- 1 — B** — Tooth eruption creates a surface where plaque can accumulate.
- 2 — A** — Frequent fermentable carbohydrates feed an acidogenic biofilm; pH falls, demineralising enamel.
- 3 — B** — Adherent gray pseudomembrane with a "bull neck" is diphtheria.
- 4 — A** — Removable white plaques with an erythematous base are pseudomembranous candidiasis.
- 5 — B** — Rheumatic fever is an immune-mediated complication involving heart, joints, skin and CNS.

SPACED REVIEW

☐ Day 1

☐ Day 3

☐ Day 7

☐ Day 21

☐ Pre-exam